Geometric Realization

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Abstract

We prove Brown–Wemyss Numerical Realization for completed Jacobi algebras with radical growth dimension at most one. The published splitting and corank reduction together leave two-variable Type E potentials with cubic part x^3. A filtered rank obstruction proves finiteness of the length-five stratum. Lifted syzygies give cubic-growth bounds for the zero-quartic stratum, and partial morsification gives the same growth exclusion for both degenerate sextic axes. These bounds hold for every higher formal tail. We construct smooth-source crepant contractions with complete local cDV targets for the surviving branches and identify their normalized basic contraction algebras. The length-five construction uses Wemyss's gluing theorem; the growth-one Type E case is realized by a divisor-to-curve contraction. Exact rational certificates establish the finite algebraic inputs.

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Contents

1 Introduction and the realization theorem **3**

2 Reduction and constructive inputs **4**
2.1 The published reduction 4
2.2 Cyclic elimination and the quartic division 4
2.3 Geometry of marked arcs 5
2.4 Three-direction potentials and inverse preparation 7
2.5 The nonzero quartic-discriminant stratum 8
2.6 Finite transfers and the length-four and length-six strata 10

3 Realization of the length-five stratum **13**
3.1 The alternating quartic and its fifth jet 13
3.2 The length-five finiteness theorem 14
3.3 The auxiliary graded algebra 15
3.4 Filtered modules with a noncentral parameter 15
3.5 Proof of length-five finiteness 17
3.6 Polynomial truncation and length-five realization 17

4 Growth exclusions **18**
4.1 Two complete-quiver growth bounds 19
4.2 Growth transfer by partial morsification 21
4.3 Proof for the degenerate sextic axes 22
4.4 PBW and syzygy lifting for the zero-quartic stratum 23
4.5 Semicontinuity for the zero-quartic stratum 25

5 Proof of the realization theorem **26**
5.1 The growth-one model 26
5.2 Completing the quartic division 26

6 Computational certificates **28**

Acknowledgements **28**

References **28**

1 Introduction and the realization theorem

The inverse problem for contraction algebras asks which completed Jacobi algebras arise from a three-dimensional crepant contraction. The Brown–Wemyss conjecture imposes only the radical growth bound $\operatorname{Jdim} \leq 1$, and includes both flopping contractions and divisor-to-curve contractions. The published reduction leaves the two-generator Type E case $x^3 + O_4$ [1, Conjecture 1.11 and Corollary 1.12].

The proof separates construction from exclusion. After the published reduction and the quartic division, Section 2 constructs marked ADE arcs for the length-three, length-four and length-six families and identifies their contraction algebras. Section 3 treats the alternating quartic: a growth bound forces the fifth coefficient, a filtered rank argument proves finiteness, and the length-five gluing theorem gives realization. The two residual branches have cubic growth. Section 4 proves their exclusion by partial morsification and by lifted syzygies followed by semicontinuity. Section 5 constructs the surviving growth-one model and completes the quartic case division.

For a finite set of generators, put

$$F_d = \mathbb{C}\langle\langle x_1, \dots, x_d \rangle\rangle, \qquad \mathfrak{m}_d = (x_1, \dots, x_d), \qquad (F_d)_{\mathrm{cyc}} = F_d/\overline{[F_d, F_d]}.$$

The bar in the cyclic quotient denotes closure of the *linear* commutator subspace. All derivative ideals below are closed two-sided ideals. For a cyclic potential φ, let

$$A_\varphi = \operatorname{Jac}(\varphi) = F_d/\overline{(\partial_{x_1}\varphi, \dots, \partial_{x_d}\varphi)}, \qquad J_\varphi = \operatorname{rad}(A_\varphi).$$

For an augmented complete local algebra A, we use

$$\operatorname{Jdim} A = \inf\left\{r \geq 0 : \dim_{\mathbb{C}}(A/J^n) \leq C_r n^r \text{ for every } n \geq 1 \text{ and some } C_r > 0\right\}.$$

When the potential is fixed, we abbreviate its Jacobi algebra to A and its radical to J. An ordinary O_q consists of words of length at least q; weighted orders will be specified separately. A necklace is a cyclic equivalence class of words. In each reduction, a good necklace avoids the stated forbidden subwords, including at the cyclic boundary; the remaining necklaces are called bad.

Definition 1.1. Write $\operatorname{Geo}(A)$ if there is a projective birational crepant morphism $\pi : X \to \operatorname{Spec} R$, where R is a three-dimensional complete local normal cDV complex algebra, X is regular, and the reduced closed fiber is $\mathbb{P}^1$, together with a continuous complete-algebra isomorphism $A \cong A_{\mathrm{con}}(\pi)$. Regularity is the smooth-source convention in this formal local setting. The generator used to define A_{con} is normalized and basic.

Conjecture 1.2 (Brown–Wemyss Numerical Realization)**.** *For every $d \geq 1$ and every cyclic formal potential $\varphi \in \mathfrak{m}_d^2$, the condition* $\operatorname{Jdim} \operatorname{Jac}(\varphi) \leq 1$ *implies* $\operatorname{Geo}(\operatorname{Jac}(\varphi))$.

Constants may be discarded. A linear term differentiates to a unit, so the nonzero augmented Jacobi algebras considered here have no linear potential term. The conjecture includes both finite-dimensional algebras and algebras of growth dimension one.

Theorem 1.3 (Geometric Realization)**.** *Let φ be a cyclic formal potential in finitely many variables, with no linear term, and let its nonzero augmented Jacobi algebra satisfy* $\operatorname{Jdim}\operatorname{Jac}(\varphi) \leq 1$*. Then* $\operatorname{Jac}(\varphi)$ *is isomorphic to the normalized basic contraction algebra of a projective birational crepant contraction with regular source, complete local normal three-dimensional cDV target, and reduced closed fiber* $\mathbb{P}^1$*. The finite-dimensional case is flopping; the growth-one case is divisor-to-curve. In residual Type E the only growth-one algebra is* $\operatorname{Jac}(x^3 + xy^3)$*.*

The proof first partitions the reduced quartic into its five orbits. The nonzero-discriminant, length-four and nonzero-pair length-six strata have complete constructive realizations. The alternating row is finite by the rank obstruction and is realized by the published length-five gluing theorem. Uniform growth excludes every remaining zero-quartic and degenerate sextic case. The final assembly also identifies the surviving growth-one Type E algebra.

2 Reduction and constructive inputs

2.1 The published reduction

Proposition 2.1 (Published reduction)**.** *Conjecture 1.2 is equivalent to its two-generator restriction*

$$\varphi = x^3 + O_4, \qquad \operatorname{Jdim}\operatorname{Jac}(\varphi) \leq 1.$$

Cases of corank at most one and two-generator cases with $\dim J^2/J^3 = 2$ *are already geometric in both growth dimensions.*

Proof. Use the splitting lemma and the corank bound in [1, Lemma 4.5 and Appendix A]. Under the growth bound, $\dim J/J^2$ is zero, one or two, and removing quadratic variables gives an isomorphic completed Jacobi algebra. Corank zero gives $\mathbb{C}$. Corank one gives $\mathbb{C}[[y]]$ or $\mathbb{C}[[y]]/(y^k)$. In corank two the cubic is nonzero and, after a linear change, has type x^3+y^3, x^2y, or x^3. The first two have second radical layer of dimension two, and the last has dimension three. The Type A and D constructions in [1, Sections 5–6 and Corollary 8.18] give exactly Definition 1.1, including divisor-to-curve models for the infinite-dimensional normal forms. These constructions establish the A/D cases in both growth dimensions. Continuous Jacobi-algebra isomorphisms preserve the realization property, so the residual implication proves the full conjecture. The converse is its specialization to two generators. □

2.2 Cyclic elimination and the quartic division

Lemma 2.2 (Removal of cyclic xx words)**.** *Every $x^3 + O_4$ is cyclically right-equivalent, by a continuous unitriangular automorphism fixing y, to a series whose higher necklaces contain no adjacent xx, including at the cyclic boundary. Its quartic part is*

$$a(xy)^2 + bxy^3 + cy^4. \tag{2.1}$$

Proof. In degree $n \geq 4$, a necklace containing xx has a representative x^2h, with h of degree $n-2$. Collect the bad part as $[x^2h_n]$ and substitute $x \mapsto x - h_n/3$. The linear change of $[x^3]$ is $-[x^2h_n]$. Multiple replacements, as well as changes to every term of degree at least four, have degree greater than n. The substitutions and their inverses stabilize on every finite word quotient. Their inverse limits give the required automorphism. The three good quartic

necklaces are precisely those in (2.1). Cyclic differentiation of a substituted word is a sum of two-sided multiples of the substituted original derivatives. Applying the inverse substitution gives both inclusions of closed derivative ideals, so all these changes preserve the completed Jacobi algebra. □

Proposition 2.3 (Five quartic orbits)**.** *Put $\Delta = b^2 - 8ac$. A linear map preserving the cubic has the form $x \mapsto \zeta x$, $y \mapsto tx + vy$, where $\zeta^3 = 1$ and $v \neq 0$. Its reduced quartic coefficients are*

$$a' = v^2(\zeta^2 a + \zeta t b + 2t^2 c), \qquad b' = v^3(\zeta b + 4tc), \qquad c' = v^4 c, \tag{2.2}$$
$$\Delta' = \zeta^2 v^6 \Delta.$$

Allowing also a nonzero scalar multiple of the potential, the five four-jet representatives are

$$x^3, \quad x^3 + (xy)^2, \quad x^3 + xy^3, \quad x^3 + y^4, \quad x^3 + (xy)^2 + y^4.$$

Their strata are respectively the zero quartic, $c = b = 0, a \neq 0$, $c = 0, b \neq 0$, $c \neq 0, \Delta = 0$, and $c \neq 0, \Delta \neq 0$.

Proof. Expand the three necklaces after the linear substitution and discard the cyclic xx terms using Lemma 2.2. In the xy^3 expansion exactly one of the three placements of the new x survives in $(xy)^2$, giving the coefficient ζtb. The y^4 contribution to $(xy)^2$ is $2t^2c$. This proves (2.2) and the discriminant identity. If $c = 0, b \neq 0$, choose $t = -\zeta a/b$ to remove a'. If $c \neq 0$, choose $t = -\zeta b/(4c)$; the remaining alternating coefficient is proportional to $a - b^2/(8c)$. Rescalings give the listed representatives. Scalar multiplication preserves the derivative ideal and hence the Jacobi algebra. □

Remark 2.4. Each quartic representative carries an arbitrary higher formal tail. The normalizations and growth bounds below treat those tails.

2.3 Geometry of marked arcs

For $\ell = 3, 4, 5, 6$, let $\mathbb{H}_\ell$ be Karmazyn's marked parameter ring, let $\mathcal{Y}_\ell \to \operatorname{Spec} R_\ell$ be the corresponding simultaneous partial resolution, and let Λ_ℓ be its specified full algebra [3, Theorem 1.3 and Section 3.3]. The free projector e_0 is retained in Λ_ℓ, and

$$C_\ell = \Lambda_\ell / \Lambda_\ell e_0 \Lambda_\ell.$$

Let $\nu : \mathbb{H}_\ell \to \mathbb{C}[[w]]$ be an arc based at the marked origin. Pull back the projective source, and then localize and complete the affine target at its central singular point. The resulting morphism is the geometry used in the construction.

Lemma 2.5 (Endomorphisms and contraction quotients)**.** *For these marked families, the full endomorphism algebra commutes with arc base change and target completion. If the resulting source is regular and the morphism is crepant with reduced closed fiber $\mathbb{P}^1$, its normalized basic contraction algebra is isomorphic to the completed e_0-quotient.*

Proof. The standard source is flat over $\mathbb{H}_\ell$, and the specified tilting bundle $E = \mathcal{O} \oplus N$ has flat full endomorphism algebra over that base. On a finite affine cover of the separated source, the Cech complex of $\mathcal{E}nd(E)$ has flat terms, no positive cohomology and flat degree-zero cohomology.

Descending induction on its syzygies shows that tensoring with any base algebra preserves this exactness. Thus ordinary endomorphisms and higher self-Ext commute with arc pullback. Generation is preserved by the affine pullback–pushforward adjunction. Target localization and completion are flat, so the same calculation applies after both operations.

To identify the correct perverse generator, use the dual $P = E^*$ of the tilting bundle E. Global generation, the relatively ample determinant and the structure sheaf summand pull back from the specified universal generator. Moreover $H^1(E) = 0$, since E is a direct summand of $\mathcal{E}nd(E)$. The criterion in [5, Propositions 3.2.6–3.2.7] therefore makes P a perversity-minus-one projective generator and E a perversity-zero generator on the pulled-back morphism. The two primitive vertex projectors then identify the normalized basic summands, each with multiplicity one.

For any central base change $H \to S$, right exactness gives

$$(\Lambda/\Lambda e_0\Lambda) \otimes_H S \cong (\Lambda \otimes_H S)/(\Lambda \otimes_H S)e_0(\Lambda \otimes_H S).$$

The full algebra after completion is finite over the complete local target. Its radical and central-maximal-ideal filtrations are cofinal: the reduction modulo the central maximal ideal is finite-dimensional, so a radical power lies in the product of the central maximal ideal with the algebra, and this product lies in the radical. Finite submodules, including the idempotent ideal, are closed. Consequently the geometric quotient is exactly the augmentation-complete quotient in the loop presentations. This endomorphism base-change argument is also developed in [Bi, Section 2.2]. □

Lemma 2.6 (The source first jet)**.** *For each residual A_{m-1} component, write its orthogonal projection polynomial as $p_m(Z, w) = Z^m + \sum_{i=0}^{m-2} a_i(w)Z^i$. The arc source is regular along the central fiber if and only if $a_0'(0) \neq 0$ at every such component.*

Proof. The completed local equation is $UV = p_m(Z, w)$ [4, Section 6]. At the central singular point its linear part is $-a_0'(0)w$. A nonzero coefficient eliminates w and gives a regular three-variable power-series ring; if it is zero the embedding dimension is four. The source is already regular at the other points of the central surface. Properness over a complete local target then extends regularity from the closed fiber: a nonempty closed nonregular locus would have image containing the closed point. □

Proposition 2.7 (Geometry of the smooth arcs)**.** *The stated first-jet condition gives a projective birational crepant contraction with regular source, complete local normal cDV hypersurface target and reduced closed fiber $\mathbb{P}^1$. In the primitive length-four, length-five, and length-six families, the contraction is small. In length three, use the quotient parameters*

$$\beta^3 = B_1\beta + B_0, \quad \gamma^3 = C_1\gamma + C_0, \quad \delta^2 = D_0, \quad \beta + \gamma + \delta = t_K/3.$$

With B_0', C_0', D_0' nonzero at zero, it is divisorial precisely when

$$t_K = 0, \qquad B_1 = C_1, \qquad B_0 = -C_0. \tag{2.3}$$

Proof. The target is a flat deformation of its central Du Val hypersurface, so it is a three-dimensional hypersurface with a Du Val Cartier section. Its central section is a domain and w is regular; factoring finite w-adic orders proves that the target is a domain. It is regular at smooth points of that section. A codimension-one singular component would meet the section

in positive dimension, contradicting its isolated singularity. The hypersurface is S_2, hence normal. Projectivity and the reduced fiber follow from the standard family.

For the three higher primitive markings, source smoothness makes all residual characteristic polynomials Eisenstein. Their Puiseux root valuations and the marked-weight calculation show that no ambient root hyperplane persists: the required divisibilities are incompatible with the allowed marked coefficients. Thus the generic surface is smooth, and the contraction is small. It is crepant since the Gorenstein target and regular source have no exceptional prime divisor.

For length three, the two cubic lists have opposite matching roots exactly under (2.3); trace zero forces $t_K = 0$. Otherwise no ambient root persists. Under those identities the generic root subsystem is three orthogonal A_1's. Their positive-valuation Puiseux traits are centered at the closed target, so their root curves remain after *the same* target completion. Their closures are exceptional divisors mapping to curves. There is no vertical exceptional divisor: the central source surface is integral and maps birationally, and only a curve is exceptional. Every horizontal discrepancy is computed on the generic surface. Its exceptional (-2)-curves have zero canonical intersection; the negative-definite intersection matrix forces every discrepancy coefficient to vanish. This proves crepancy in the divisorial case. Finally, apply Lemma 2.5 for the contraction algebra. The root and partial-resolution inputs come from [4, Sections 5–6], and the explicit marked coordinates come from [3, Examples 4.24–4.27]. □

Remark 2.8. The quotient of a flat full algebra need not be flat. For example, the $\mathbb{C}[[w]]$-free order $\left(\begin{smallmatrix} \mathbb{C}[[w]] & \mathbb{C}[[w]] \\ w\mathbb{C}[[w]] & \mathbb{C}[[w]] \end{smallmatrix}\right)$ has free-vertex quotient $\mathbb{C}[[w]]/(w)$. Likewise, a nonzero finite local algebra on which a radical central element acts nilpotently cannot be flat over its parameter DVR. Accordingly, the construction pulls back the full endomorphism algebra before forming its idempotent quotient.

2.4 Three-direction potentials and inverse preparation

Lemma 2.9 (Common central value)**.** *Let $P, Q, R \in u\mathbb{C}[[u]]$, and let U, V, W be their zero-constant antiderivatives. The completed Jacobi algebra of $F = U(x) + V(y) + W(-x-y)$ is the complete algebra with a central generator w and relations*

$$P(x) = Q(y) = R(z) = w, \qquad x + y + z = 0.$$

Proof. The two cyclic derivatives are $P(x) - R(-x-y)$ and $Q(y) - R(-x-y)$. Their common value commutes with x because it is a series in x, and with y because it is a series in y. It is therefore central. Adjoin $z = -x - y$ and $w = P(x)$; elimination gives mutually inverse continuous augmented maps. Completion is taken in all the augmented generators. □

Lemma 2.10 (One-variable preparation and its first jet)**.** *If $H(u) = h_d u^d + O(u^{d+1})$, $h_d \neq 0$, then $H(u) - w$ is a unit times a unique monic degree-d polynomial $M_H(u, w)$. Its lower coefficients lie in $w\mathbb{C}[[w]]$. After the shift $v = u + \rho_H(w)$, $\rho_H = m_{d-1}/d$, write*

$$M_H(v - \rho_H, w) = v^d - \sum_{i=0}^{d-2} T_{H,i}(w) v^i.$$

Then $T'_{H,0}(0) = 1/h_d$. If $\widetilde{H} - H \in u^M\mathbb{C}[[u]]$, $M \geq d+1$, then

$$\begin{aligned} \widetilde{\rho}_H - \rho_H &\in w^{\lfloor M/d \rfloor}\mathbb{C}[[w]], \\ \widetilde{T}_{H,i} - T_{H,i} &\in w^{\lceil (M-i)/d \rceil}\mathbb{C}[[w]]. \end{aligned} \tag{2.4}$$

Proof. Formal preparation applies in $\mathbb{C}[[w]][[u]]$. Equivalently, put $w = q^d$ and take the d inverse branches $u_j = \zeta_d^j h_d^{-1/d} q + O(q^2)$. Their product polynomial is invariant under $q \mapsto \zeta_d q$ and lies in $\mathbb{C}[[w]][u]$. Its constant coefficient is $-w/h_d + O(w^2)$; depression changes that coefficient only by $O(w^2)$. An u^M change moves each inverse branch by order at least q^{M-d+1}. Taking the elementary symmetric functions gives the indicated coefficient bounds; depression preserves them. In particular, a potential correction of order $dN+1$ changes every prepared arc coefficient only by w^N. The estimates hold at every complex coefficient point. □

Theorem 2.11 (Realization of the three-direction families)**.** *Suppose the exact orders of P, Q, R are one of*

$$(3,3,2), \qquad (2,4,3), \qquad (2,3,5),$$

with nonzero leading coefficients. The Jacobi algebra of $U(x)+V(y)+W(-x-y)$ is geometric. The marked lengths are respectively three, four and six. The latter two are small and finite-dimensional. In length three the divisorial alternative is given by (2.3) on the prepared arc.

Proof. Use Lemma 2.9, prepare the three equations $P(x)-w, Q(y)-w, R(z)-w$, and depress each polynomial by its central shift. The shifted sum is $\beta+\gamma+\delta = \rho_P+\rho_Q+\rho_R$; set $t_K = \ell(\rho_P+\rho_Q+\rho_R)$. The remaining coefficients are exactly Karmazyn's quotient parameters. Substitution of the prepared unit into a complete augmented algebra is valid and preserves its closed ideal. Inverse shifts recover all original generators. Lemma 2.10 makes every residual constant derivative nonzero, so Proposition 2.7 constructs the geometry and Lemma 2.5 identifies its contraction algebra. A small quotient is finite over the completed target and supported at its closed point, hence has finite complex dimension. □

2.5 The nonzero quartic-discriminant stratum

Lemma 2.12 (Two cyclic tangent calculations)**.** *Give x, y weights $3, 2$, respectively, and set $r_x = 3x^2+y^3$, $S = xy^2+yxy+y^2x$. Every cyclic homogeneous polynomial of weight $d > 9$ has the form*

$$[ur_x + vS] + \lambda[y^{d/2}], \qquad \mathrm{wt}(u) = d-6, \quad \mathrm{wt}(v) = d-7,$$

where the last summand occurs only for even d. For $\lambda_0 \neq 0$, the ordinary graded quotient

$$D_{\lambda_0} = \mathbb{C}\langle x, y\rangle/(x^2, S+\lambda_0 y^3, [x, y^3])$$

has cyclic quotient spanned by $1, x, xy$ and pure powers of y. In degrees at least seven the pure classes vanish unless the degree is divisible by three. If $L_b = y+bx$ and $t_b = 1-\lambda_0 b$, then

$$[L_b^{3k}] = t_b^k [y^{3k}], \qquad [L_b^5] = \frac{1+5t_b^2}{6}[y^5]. \tag{2.5}$$

Proof. For the first assertion, $x^2 = -y^3/3$ is central in the quotient by r_x, S. Over that central coefficient, the reductions of $xx, yyy, yyx, yxyx$ leave the twelve spanning words

$$1, y, y^2, x, xy, xy^2, xyx, xyxy, xyxy^2, yx, yxy, yxy^2.$$

Rotating their traces and multiplying the relation $S=0$ reduces them to pure y powers and x, xy; the latter have weights at most five. Homogeneous two-sided relation multiples rotate to

$[ur_x + vS]$, with the asserted coefficient weights. These spanning expressions give the required decomposition.

In the second quotient put $Z = y^3$ and $q = xyx$. The defining commutator makes Z central, and $x(S + \lambda_0 Z) - (S + \lambda_0 Z)x = 0$ gives $xyxy = yxyx$. Thus q is central and $q^2 = 0$. The same twelve words span over $\mathbb{C}[Z]$. Traces of $S + \lambda_0 Z$ multiplied by $1, y, y^2, x, yx$ give

$$[xy^2] = -\lambda_0[Z]/3, \qquad Z[x] = -\lambda_0 Z[y]/3, \qquad Z[xy] = -\lambda_0 Z[y^2]/3,$$
$$q[y] = (\lambda_0^2/3)Z[y], \qquad q[y^2] = (\lambda_0^2/6)Z[y^2].$$

These identities reduce all twelve traces. Multiplying the last two identities by q kills $Z^2[y]$ and $Z^2[y^2]$. Finally, $L_b^3 = t_b Z + b^2 q$; expand its powers and use $[q] = 0$. Multiplying this identity by L_b^2 and using the displayed traces proves the fifth-power formula. Homogeneity gives, for an ordinary degree $n \geq 5$ discrepancy, a decomposition $[\mu y^n + ux^2 + v(S + \lambda_0 y^3) + k[x, y^3]]$, with coefficient degrees $n-2, n-3, n-4$, respectively. □

Theorem 2.13 (Complete tails for $\Delta \neq 0$)**.** *Let $f = x^3 + O_4$ have reduced quartic $a_4(xy)^2 + b_4 xy^3 + c_4 y^4$. If $b_4^2 - 8a_4c_4 \neq 0$, then* $\mathrm{Geo}(\mathrm{Jac}(f))$*. The construction is a marked length-three E_6 arc. If $c_4 \neq 0$, it is small. If $c_4 = 0$, there is a cyclic right-equivalence to*

$$x^3 + xy^3 + p(y), \qquad p \in y^5\mathbb{C}[[y]],$$

and the construction is divisorial precisely for $p = 0$. All other members are finite-dimensional. The sole infinite-dimensional Jacobi algebra in this stratum, up to complete-algebra isomorphism, is $\mathrm{Jac}(x^3 + xy^3)$*, of growth dimension one.*

Proof. **The branch $c_4 = 0$.** If $c_4 = 0$, shear and rescale to make the quartic xy^3. After cyclic xx elimination, all remaining terms have weight greater than nine for $\mathrm{wt}(x, y) = (3, 2)$. The first calculation in Lemma 2.12, applied in increasing weight, removes every term except the possible pure powers. At weight d the corrections $x \mapsto x - u$, $y \mapsto y - v$ raise generator weights by $d - 9$; two replacements have weight at least $9 + 2(d - 9) > d$. The corrections and their inverses converge, giving the stated p.

Put $\beta = y + x/2$, $\gamma = -y + x/2$, and consider

$$G_h = x^3 + \beta^4/4 - \gamma^4/4 + h(\beta), \qquad h \in u^5\mathbb{C}[[u]].$$

Its base cyclic class is $x^3 + xy^3 + x^3y/4$, with correction weight eleven. At each even weight $2m$, the coefficient of u^m in h contributes exactly the remaining y^m class. At odd weights, only tangent terms remain. The same increasing weight corrections construct h and a convergent automorphism sending G_h to the prescribed potential. In the generators β, γ, the third term is $-(-\beta - \gamma)^3$. The derivative orders are $(3, 3, 2)$, so Theorem 2.11 supplies the geometry.

The branch $c_4 \neq 0$. If $c_4 \neq 0$, choose τ with $a_4 + b_4\tau + 2c_4\tau^2 = 0$. The shear kills the alternating quartic, while $(b_4 + 4c_4\tau)^2 = \Delta \neq 0$. After rescaling and removing new xx quartics, the potential is $f_c = x^3 + xy^3 + cy^4 + O_5$, with $c \neq 0$. Choose a root r of

$$4cr(1-r) + 2r - 1 = 0, \quad L_1 = y + rx, \quad L_2 = -y + (1-r)x, \quad a = c(1-r) + 1/4, \quad d = cr - 1/4.$$

Here $r \neq 0, 1$ and $a, d \neq 0$. The identities $a + d = c$, $4(ar - d(1-r)) = 1$ and $ar^2 + d(1-r)^2 = 0$ show that $aL_1^4 + dL_2^4$ has exactly the desired reduced quartic. Its other quartics are removed by a quadratic x correction.

Take $\lambda_0 = 4c$ in the second tangent calculation and put $t = 1 - 4cr = r/(1-r) \neq 0$. The powers of L_1 supply every degree $3k$ obstruction with coefficient t^k. In degree five, the coefficients of L_1^5, L_2^5 are $(1+5t^2)/6, -(1+5t^{-2})/6$, which cannot both vanish. Thus one-variable corrections $h_1(L_1), h_2(L_2)$, each starting in degree five, supply every residual class. Gradient corrections remove $[ux^2 + v(S + 4cy^3)]$. To remove $[k[x, y^3]]$, use the derivation $D(x) = [k, x], D(y) = 0$ and its exponential. It preserves the cyclic cubic exactly, raises degree by $n-4$, and its second variation of the quartic has degree $2n-4 > n$. Consequently, the degree induction is valid for arbitrary complex coefficients, and the maps and their inverses converge. It yields

$$x^3 + aL_1^4 + dL_2^4 + h_1(L_1) + h_2(L_2).$$

Since $L_1 + L_2 = x$, this again has derivative orders $(3, 3, 2)$. The prepared cubic constant derivatives are $1/(4a)$ and $1/(4d)$; their sum is $c/(4ad) \neq 0$. Thus the divisorial identity $B_0 = -C_0$ fails already to first order, and the contraction is small.

Finite dimension and growth one. An intrinsic check distinguishes the finite and infinite cases in the pure-series row. Put $T = x^2 = -y^3/3$, a central element. The twelve-word reduction makes $A_p = \mathrm{Jac}(x^3 + xy^3 + p(y))$ finite over $\mathbb{C}[[T]]$. For $p = 0$ the critical overlaps reduce to the two derivative relations, with no further relation among those twelve words; this gives a free rank-twelve module. The positive weights compare the central and radical filtrations of A_p, showing that its radical growth is linear. Suppose $p \neq 0$ but the generic algebra is nonzero. Over an algebraic closure of $\mathbb{C}((T))$, it has a simple module of dimension $r \leq 3$, since its full matrix image has dimension $r^2 \leq 12$. On this module $X^2 = T$, $Y^3 = -3T$. Choose $t^3 = T$ and $\lambda_i = \zeta_i t$, $\zeta_i^3 = -3$. The diagonal block of the second derivative relation gives $X_{ii} = -p'(\lambda_i)/(3\lambda_i^2)$. Thus $\operatorname{tr} X \in \mathbb{C}((t))$, but it is also $\ell\sqrt{T}$. The square root has nonintegral t-valuation, so $\ell = 0$ and $r = 2$. If $p = h_n y^n + \cdots$, the two selected diagonal traces have nonzero leading sum proportional to $2\zeta_i^{n-3}$ or $\zeta_i^{n-3}(1 + \omega^{n-3})$, $\omega^3 = 1$. This contradicts trace zero. The generic algebra therefore vanishes; finite central generation makes A_p torsion of bounded exponent and finite-dimensional. The marked root criterion identifies the $p = 0$ arc as divisorial. For $p \neq 0$, a divisorial contraction would have nonzero quotient support along its exceptional image curve, contrary to finite length. This proves the alternatives. □

2.6 Finite transfers and the length-four and length-six strata

The next lemma integrates the tangent directions for arbitrary complex coefficients. Its order estimates control the simultaneous limits of both the family and the coordinate maps.

Lemma 2.14 (Positive-order family transfer)**.** *Let $G = U(x) + V(y) + W(L)$, $L = y - \lambda x$, $\lambda \neq 0$. Suppose U' has exact order $d \geq 2$ and $W'(u) = bu^e + O(u^{e+1})$, $e > d$, $b \neq 0$. For $k \geq 1, s \geq 0$ with $r = d(k-1) + s \geq 2$ and for every $\rho \in \mathbb{C}$, there exist a modified U_ρ and an identity-tangent automorphism Ψ_ρ fixing y, such that*

$$\begin{aligned} \operatorname{ord}(U_\rho - U) &\geq dk + s, \\ \operatorname{ord}(\Psi_\rho(x) - x),\ \operatorname{ord}(\Psi_\rho^{-1}(x) - x) &\geq r, \\ [\Psi_\rho(U_\rho(x) + V(y) + W(L))] &= [G] + \rho b^k [L^{ek} x^s] + O_{ek+s+1}. \end{aligned}$$

The y-version uses V' of order d, fixes x, and adds $\rho b^k [L^{ek} y^s]$ in degree $ek + s$.

Proof. Introduce an auxiliary time τ, set $A_\tau = U'_\tau(x)/\lambda$ and $B = W'(L)$, and solve

$$\partial_\tau U_\tau(u) = \rho u^s (U'_\tau(u)/\lambda)^k,$$
$$C_\tau = \sum_{i=0}^{k-1} B^{k-1-i} x^s A_\tau^i, \qquad D_\tau(x) = -\rho C_\tau/\lambda, \quad D_\tau(y) = 0.$$

The coefficient recursion raises the index by $r - 1 > 0$, so each coefficient of U_τ is polynomial in τ. Since C_τ has order at least r, the same triangular recursion solves $\partial_\tau \Psi_\tau = \Psi_\tau D_\tau$, $\Psi_0 = \mathrm{id}$, with the coefficients of Ψ_τ polynomial in τ. On each finite word quotient, this gives an invertible unipotent substitution whose inverse has polynomial coefficients and the same order bound. Therefore, evaluating coefficientwise at $\tau = 1$ is valid.

Noncommutative telescoping and cyclic rotation give

$$[x^s A_\tau^k] - [B^k x^s] = [C_\tau(A_\tau - B)], \quad \partial_\tau[\Psi_\tau(G_\tau)] = \rho[\Psi_\tau(B^k x^s)].$$

The right side has leading degree $ek + s$, independent of time. Integrating its polynomial coefficients proves the assertion. For the other branch, use $A_\tau = -V'_\tau(y)$, $\partial_\tau V_\tau = \rho u^s(-V'_\tau)^k$, $C_\tau = \sum B^{k-1-i} y^s A_\tau^i$ and $D_\tau(y) = \rho C_\tau$. The exact flow identity gives the asserted leading term and order bounds. □

Lemma 2.15 (The weighted y^4 reduction)**.** *Every $x^3 + y^4 + O_5$ is right-equivalent, with the entire tail retained, to*

$$x^3 + y^4 + \alpha_{17} xyxy^2 + \alpha_{20}(xy^2)^2 + \alpha_{21}(xy)^3 + O_{\mathrm{wt}}(24), \qquad \mathrm{wt}(x, y) = (4, 3),$$

where every remaining necklace avoids the cyclic subwords xx and yyy. The corresponding ordinary remainder has order at least seven.

Proof. In weight $n > 12$, rotate each bad necklace into $x^2 u_n$ or $y^3 v_n$. The substitutions $x \mapsto x - u_n/3$, $y \mapsto y - v_n/4$ raise generator weights by $n - 12$, remove the bad part, and create nonlinear terms only above weight n. Both substitutions and their inverses converge. Steps in weights thirteen and fourteen are the identity because the initial tail has ordinary order at least five, so no $(xy)^2$ term is introduced. A remaining necklace is an alternating sequence of blocks xy and xy^2, with at least two x's. The only remaining possibilities below weight twenty-four are the displayed words of weights seventeen, twenty, and twenty-one. All further possibilities have ordinary length at least seven. This proves the normalization. □

Theorem 2.16 (Length-four realization)**.** *For every $a \neq 0$ and every cyclic formal tail $h \in O_6$,*

$$f = x^3 + y^4 + a\, xyxy^2 + h$$

is right-equivalent to $x^3 + V(y) + W(y - x)$, where $V(u) = u^4 + O(u^5)$, $W(u) = (a/5)u^5 + O(u^6)$. Its Jacobi algebra is finite-dimensional and geometric through a smooth marked length-four small contraction.

Proof. **Residual cyclic classes.** Put $u = xy, v = xy^2, g = xyxy^2$ and $q_{17} = [x, \partial_x g]$. Consider the graded algebra

$$D_{17} = \mathbb{C}\langle x, y\rangle/(x^2, y^3, q_{17}).$$

In its cyclic quotient, the degree $m \geq 6$ residuals are $u^{m/2}$ for even m, and additionally $u^{2k-2}v^2$ for $m = 4k+2$; odd degrees vanish. For this homogeneous calculation, reduce words avoiding xx, yyy to alternating words in u, v, and use cyclic rotations of the conjugation relation to move the two v's together. The relations eliminate the odd-degree classes and leave the displayed congruence classes. Every error therefore decomposes into these residuals and $[Hq_{17} + x^2u_m + y^3v_m]$; the coefficient degrees are $m-5, m-2, m-3$, respectively. Spanning suffices for the induction.

Matching degree six. The seed $x^3 + y^4 + (a/5)(y-x)^5$ has only one good necklace of degree five, namely ag. Remove its bad fifth necklaces by degree-three x and degree-two y corrections. The next step is a finite degree-six calculation: change the family by $Ay^5 + B(y-x)^6$, and compensate with $y \mapsto y - Ay^2/4$. Modulo bad necklaces the change is

$$(-aA/4 + 3B)v^2 - 2Bu^3. \tag{2.6}$$

The term of degree six has three v^2 rotations and two u^3 rotations; the compensation on ag supplies $-aAv^2/4$. All nonlinear degree-six effects on the pure quartic and fifth powers are bad. The two-coefficient determinant is $a/2 \neq 0$, so this matches every degree-six residual at arbitrary complex coefficients.

Higher degrees and convergence. Suppose agreement holds below degree $m \geq 7$. For $m = 4k$, changing W by μu^m supplies $2\mu u^{2k}$. For $m = 4k+2$ with $k \geq 2$, put $A_y = -V'(y)$, $B_y = W'(y-x)$, and make the following changes:

$$V \mapsto V + \lambda y^2 A_y^k, \qquad W \mapsto W + \mu u^m,$$
$$y \mapsto y + \lambda \sum_{i=0}^{k-1} B_y^{k-1-i} y^2 A_y^i.$$

Telescoping gives residual vectors $\lambda a^k u^{2k-2}v^2$ and $\mu(-2u^{2k+1} + c_k u^{2k-2}v^2)$, which span as the two update parameters vary, since $a^k, -2 \neq 0$. The family and coordinate changes start in orders $3k+2, 3k-1$; nonlinear errors have order at least $6k > 4k+2$. In odd degrees only the kernel correction is needed. At every step, the operations use the current V, W.

Conjugate x by $\exp(H/a)$ to add $[Hq_{17}]$. This conjugation preserves the cyclic cubic and quartic exactly; its second effect on ag has degree $2m-5 > m$. Principal corrections $x \mapsto x + u_m/3$, $y \mapsto y + v_m/4$ remove the other kernel terms without changing earlier degrees. Because the orders of the family updates, maps, and inverse maps tend to infinity, these operations stabilize on every finite word quotient. The resulting limits give the asserted right-equivalence. With $X = x, Y = y - x$, the derivative orders are $(2, 4, 3)$; apply Theorem 2.11. □

Theorem 2.17 (Nonzero sextic-pair realization)**.** *For $b, c \neq 0$, every*

$$f = x^3 + y^4 + b(xy^2)^2 + c(xy)^3 + O_7$$

is right-equivalent to $x^3 + V(y) + W(y - \lambda x)$, where

$$\lambda = -3c/(2b), \quad d_0 = b/(3\lambda^2), \quad V(u) = u^4 + O(u^5), \quad W(u) = d_0u^6 + O(u^7).$$

It is realized by a smooth marked length-six small contraction and has finite-dimensional Jacobi algebra.

Proof. Put $L = y - \lambda x$, $g_6 = b(xy^2)^2 + c(xy)^3$, $q_6 = [x, \partial_x g_6]$. In $D_6 = \mathbb{C}\langle x, y\rangle/(x^2, y^3, q_6)$, L^5 is central and its cocenter is generated over $\mathbb{C}[L^5]$ by $1, y, y^2, L, L^2, L^3, L^4$. The homogeneous alternating-block reduction uses $3d_0\lambda^2 = b, -2d_0\lambda^3 = c$. Hence the residuals in degree $m \geq 7$ are L^m, together with $L^{5k}y^j$ when $m = 5k + j$, $j = 1, 2$. The kernel is $[x^2u_m + y^3v_m + H_m q_6]$, with coefficient degrees $m - 2, m - 3, m - 6$.

The seed $x^3 + y^4 + d_0L^6$ has the good coefficients $3d_0\lambda^2, -2d_0\lambda^3$, which are exactly b, c. Remove its bad degree-six necklaces by principal corrections. If the current family agrees through degree $m - 1$, a change of degree m in W supplies L^m. For $L^{5k}y^j$, apply Lemma 2.14 in the y-branch with $d = 3, e = 5, s = j$, and divide the amplitude by $(6d_0)^k$. The correction order $3(k - 1) + j$ is at least two: the first possible k is two for $j = 1$ and one for $j = 2$. The exact flow handles these low-order cases and changes V only from order $3k + j$.

Conjugating x by $\exp(H_m)$ adds $[H_m q_6]$; the second effect has degree $2m - 6 > m$, and principal corrections remove the remaining kernel. These preserve all earlier degrees. The orders of the family updates, coordinate maps, and inverse maps tend to infinity with m, so all three sequences converge simultaneously. With $X = \lambda x, Y = -y$, the derivative orders are $(2, 3, 5)$, and Theorem 2.11 supplies the geometry. □

3 Realization of the length-five stratum

Under the bound Jdim ≤ 1, the alternating-quartic branch must have a nonzero pure fifth coefficient. After normalizing that coefficient, a filtered rank argument proves finiteness for every remaining formal tail. Polynomial gluing then realizes the same completed Jacobi algebra.

3.1 The alternating quartic and its fifth jet

Theorem 3.1 (A uniform weighted growth obstruction)**.** *If every word of f has weight at least six for* $\mathrm{wt}(x, y) = (2, 1)$*, then for every $N \geq 1$*

$$\dim_{\mathbb{C}}\big(\mathrm{Jac}(f)/J^N\big) \geq \#\{(a, b, c) \in \mathbb{Z}^3_{\geq 0} : 2a + b + 3c < N\} \geq \big(\lfloor (N - 1)/6\rfloor + 1\big)^3. \tag{3.1}$$

In particular $\mathrm{Jdim}\,\mathrm{Jac}(f) \geq 3$.

Proof. Fix the weighted cutoff K_N. Only potential terms of weight at most $N + 1$ affect derivatives below that cutoff. Parametrize all those cyclic coefficients near $f_0 = (xy - yx)^2/2$ by independent commuting parameters e_i, write $g = \sum e_i h_i$, and use the coefficient ring $T = \mathbb{C}[[e_1, \ldots, e_r]]$. Adjoin a generator ζ of weight three and use the potential $\zeta(xy - yx) - \zeta^2/2 + g$. Its derivative relations give the rewriting rules

$$yx \longmapsto xy - \zeta, \quad \zeta y \longmapsto y\zeta + \partial_x g, \quad \zeta x \longmapsto x\zeta - \partial_y g.$$

At a finite parameter cutoff, derivative corrections raise parameter degree; the remaining rules decrease word length or inversion number with respect to $x < y < \zeta$. These rules terminate and never lower weight. The sole overlapping leading word is ζyx. Their difference is $-[y, \partial_y g] - [x, \partial_x g] = 0$ by the free cyclic identity. Disjoint replacements commute. Induction on the decreasing measure therefore gives unique ordered normal forms $x^a y^b \zeta^c$, $2a + b + 3c < N$. Passing to the compatible parameter cutoffs gives a free finite T-module of that rank. Eliminating $\zeta = xy - yx$ identifies this module with the weighted cutoff of the original two-generator Jacobi family.

The finite relation matrix has polynomial coefficient entries. Its generic rank is the word-space dimension minus the displayed rank; every larger minor is identically zero. At every complex coefficient specialization its rank can only decrease, so the quotient dimension can only increase. Translating by f_0 parametrizes all potential coefficients of weighted order at least six. This proves the same lower bound for every specialization, including specializations far from f_0. For a complete potential, its truncated derivative ideal has the same image in the finite quotient, so passage to closure changes nothing there. Finally $J^N \subseteq K_N \operatorname{Jac}(f)$, giving the first inequality. All triples with entries at most $\lfloor (N-1)/6 \rfloor$ satisfy the weight condition, giving the second. □

Corollary 3.2 (The alternating-quartic five-jet)**.** *If* $f = x^3/3+(xy)^2/2+O_5$ *and* $\operatorname{Jdim}\operatorname{Jac}(f) \leq 1$*, its pure* y^5 *coefficient is nonzero. A formal coordinate change and a nonzero scalar multiple put its five-jet into*

$$x^3/3 + (xy)^2/2 - xy^4 - y^5/5.$$

Proof. The only good fifth necklaces are $xyxy^2, xy^4, y^5$. If the last coefficient vanishes, every word has $(2,1)$-weight at least six, contradicting Theorem 3.1. Write the coefficients as A, B, C, $C \neq 0$. Scale $x \mapsto s^2x, y \mapsto sy$ and scale the potential by s^{-6}, with $s = -5C$, to make $C = -1/5$. Shear $y \mapsto y + dx$, then restore the four-jet using the correction $x \mapsto x - dxy - d^2x^2/2$. The shear changes the good one-x fifth coefficient to $B - d$, while the restoring correction changes only necklaces with at least two x's. Choose $d = B + 1$, remove bad fifth necklaces with a cubic x correction, and finally use $y \mapsto y - A'y^2$ to cancel the remaining $A'xyxy^2$. All resulting terms of degree at least six remain unrestricted. □

3.2 The length-five finiteness theorem

Let

$$p(x) = a_4x^4 + O(x^5), \qquad a_4 \neq 0, \qquad I_p' = p, \quad I_p(0) = 0,$$

and put

$$F_p = \frac{y^3}{3} + y^2x^2 + \frac{yxyx}{2} + yx^4 + \frac{x^6}{6} - I_p(x).$$

Theorem 3.3. *For every cyclic formal series* $h \in O_6$*, the algebra*

$$A = \operatorname{Jac}(F_p + h)$$

is finite-dimensional over $\mathbb{C}$*.*

The finiteness proof is a divisibility contradiction. If the Jacobi algebra were infinite-dimensional, its associated graded algebra would have generic rank 25 over the central polynomial ring of an auxiliary algebra. The leading fourth-power relation would force that rank to be divisible by 4. The filtered lemmas below apply to a lift of the graded central element without assuming that the lift is central in the unknown algebra. Proposition 3.9 then gives geometric realization.

3.3 The auxiliary graded algebra

Give x, y weights $1, 2$. Set

$$Q = y^2 + yx^2 + x^2y + xyx + x^4, \qquad S = yxy + y^2x + yx^3, \qquad B = \mathbb{C}\langle x, y\rangle/(x^4, Q).$$

The free polynomial identities

$$[S, y] = y[x, Q], \qquad [S, x] = [y, Q] + x[x, Q]$$

show that S is central in B; it is homogeneous of weight five.

Lemma 3.4. *The algebra B is finite over $\mathbb{C}[S]$, and*

$$\dim_{\mathbb{C}} B/(S) = 40, \qquad B/(S-1) \cong M_5(\mathbb{C}).$$

Consequently $B \otimes_{\mathbb{C}[S]} \mathbb{C}(S)$ is central simple of dimension 25.

Proof. The companion script `nr_B5_generic_certificate.py` uses exact rational arithmetic and a terminating weighted degree-lexicographic order. It completes the two defining relations of B after resolving all overlap and inclusion ambiguities. Adding S gives exactly 40 irreducible words. Adding $S - 1$ gives exactly 25 irreducible words. Both specializations are taken in the ordinary polynomial algebra.

For the second quotient, the 25-dimensional regular trace pairing has nonzero determinant

$$298023223876953125 = 5^{25},$$

and its center has dimension one. In characteristic zero, these two calculations show that the second quotient is a simple matrix algebra and hence isomorphic to $M_5(\mathbb{C})$. The full Groebner bases, basis words, and exact outputs are recorded in `nr_B5_generic_certificate.json`.

Finite-dimensionality of $B/(S)$ implies finite generation of B over $\mathbb{C}[S]$: lift homogeneous spanning words and induct on weight, subtracting multiples of S. Finally adjoin z with $z^5 = S$ to $\mathbb{C}(S)$. Rescaling $x = zX$, $y = z^2Y$ identifies the generic algebra after this faithfully flat field extension with the scalar extension of $B/(S-1)$. Central simplicity and its dimension descend. □

3.4 Filtered modules with a noncentral parameter

Let A be a separated complete algebra with a decreasing positive integer filtration F^nA, $F^0A = A$, finite-dimensional associated graded pieces, and $F^iAF^jA \subseteq F^{i+j}A$.

Lemma 3.5. *Suppose $\operatorname{gr} A$ is finite over the polynomial ring $\mathbb{C}[\sigma]$, where σ is central and homogeneous of degree $e > 0$. Choose a lift $s \in F^eA$ of σ. Regard A as a left $D = \mathbb{C}[[t]]$-module by $t \cdot a = sa$. Then A is a finite D-module and*

$$\operatorname{rank}_D A = \operatorname{rank}_{\mathbb{C}[\sigma]} \operatorname{gr} A.$$

Its D-torsion submodule is a right ideal and is finite-dimensional over $\mathbb{C}$.

Proof. Lift a finite homogeneous set of $\mathbb{C}[\sigma]$-module generators of $\operatorname{gr} A$. Successively eliminate initial forms by left powers of s. Completeness allows the resulting coefficients to sum to power series in s. This proves finite generation over D. If the largest generator weight is c, the same reduction gives, after increasing c if necessary,

$$F^{en+c}A \subseteq s^n A \subseteq F^{en}A.$$

Indeed an element whose filtration degree is at least $en + c$ is reduced using only powers s^j with $j \geq n$; the infinite reduction converges. For a finite graded $\mathbb{C}[\sigma]$-module of rank r, its cumulative Hilbert function is $rN/e + O(1)$. The inclusions above show

$$\dim_{\mathbb{C}} A/s^n A = rn + O(1).$$

The structure theorem over D identifies the leading coefficient in the preceding dimension estimate with $\operatorname{rank}_D A$. Right multiplication commutes with left multiplication by s, so it preserves the torsion submodule. Finite generation over the DVR makes this submodule finite-dimensional over its residue field. □

Lemma 3.6 (Filtered determinant test)**.** *In the situation of Lemma 3.5, assume the common rank is $r > 0$. Suppose $z \in F^1 A$, $m \geq 1$, $c \in \mathbb{C}^\times$, and a one-variable formal series v satisfy*

$$q(z) = cz^m(1 + zv(z)) = s + \epsilon, \qquad \epsilon \in F^{e+1}A.$$

Then m divides r.

Proof. Let T be the D-torsion submodule and $M = A/T$, free of rank r. Give M the quotient filtration. Right multiplications R_a are D-linear. Since the initial form of s is central,

$$(R_s - t\,\mathrm{id})(F^n M) \subseteq F^{n+e+1}M.$$

Thus $R_{q(z)} = t\,\mathrm{id} + E$ with E increasing filtration by at least $e + 1$.

Set $t = u^e$ and work over $\mathbb{C}((u))$. The submodule

$$\mathcal{L} = \sum_{n \geq 0} u^{-n}\big(F^n M \otimes_D \mathbb{C}[[u]]\big) \subset M \otimes_D \mathbb{C}((u))$$

is a full $\mathbb{C}[[u]]$-lattice. The good-filtration bounds in Lemma 3.5 bound this sum inside a fixed negative power of u times the standard lattice, and the $n = 0$ term contains that lattice. An operator increasing filtration by j maps $\mathcal{L}$ into $u^j\mathcal{L}$. Consequently

$$R_{q(z)} = u^e(I + uC) \quad \text{on } \mathcal{L}, \qquad v_t(\det R_{q(z)}) = r.$$

The operator R_z sends $\mathcal{L}$ into $u\mathcal{L}$, so $R_{1+zv(z)}$ has unit determinant. The polynomial factorization of q therefore gives

$$r = m\, v_t(\det R_z).$$

The determinant is nonzero and belongs to $\mathbb{C}((t))$, so its t-valuation is an integer. This proves the divisibility. □

Remark 3.7. Left multiplication by s supplies the D-module structure, while right multiplication supplies the D-linear determinant operators.

3.5 Proof of length-five finiteness

Proof of Theorem 3.3. Split $h = g(x) + h_{\mathrm{mix}}$, with $g \in x^6\mathbb{C}[[x]]$ and every word of h_{mix} containing y. Since ordinary length is at least six, every such mixed word has weight at least seven. Put $\widetilde{p} = p - g'$, whose leading term is still a_4x^4. Direct cyclic differentiation gives

$$\partial_y F_p = Q, \qquad \partial_x F_p = S + xQ - p(x).$$

Hence the two relations in A imply

$$Q \in F^5A, \qquad \widetilde{p}(x) = S + \big(\partial_x h_{\mathrm{mix}} - x\partial_y h_{\mathrm{mix}}\big),$$

where the expression in parentheses has weight at least six. The initial relations are x^4 and Q; in particular there is a graded surjection

$$B \twoheadrightarrow \operatorname{gr} A.$$

The associated graded is therefore finite over the image of $\mathbb{C}[S]$.

Suppose A were infinite-dimensional. Then $\operatorname{gr} A$ is infinite-dimensional and has nonzero generic fiber over $\mathbb{C}[S]$. This generic fiber is a nonzero quotient algebra of the central simple algebra of Lemma 3.4, so it must be the entire algebra. Its rank is exactly 25.

Apply Lemma 3.5 to the lift $s = S(x, y) \in A$. Apply Lemma 3.6 with $e = 5$, $z = x$, $q = \widetilde{p}$, and $m = 4$. It would follow that 4 divides 25, a contradiction. Thus A is finite-dimensional. □

3.6 Polynomial truncation and length-five realization

Lemma 3.8. *If* $\operatorname{Jac}(f)$ *has radical nilpotence exponent* L*, then replacing* f *by its ordinary truncation through any degree* $N \geq L + 1$ *leaves its closed Jacobi ideal unchanged.*

Proof. Let I and I_N denote the two ideals. Their generators agree modulo $\mathfrak{m}^N$, and $\mathfrak{m}^L \subseteq I$. Therefore

$$\mathfrak{m}^L \subseteq I_N + \mathfrak{m}^{L+1}.$$

Multiplication and iteration give this containment with $\mathfrak{m}^{L+1}$ replaced by arbitrarily high powers. Closedness gives $\mathfrak{m}^L \subseteq I_N$. The differences of corresponding derivatives now belong to both ideals, so the ideals are equal. □

For a two-series potential, write $G = F_p + g(x) + \ell(x)y$, where $g \in x^6\mathbb{C}[[x]]$ and $\ell \in x^5\mathbb{C}[[x]]$, and set $\widetilde{p} = p - g'$. Brown–Wemyss's free-necklace construction gives the following explicit smooth CY neighborhood on two charts:

$$b = a^{-1}, \quad U = a^{-1}u, \quad V = a^3v + a^{-1}u^2 + u^3 + au^4 + a^2(u^5 - \widetilde{p}(u)) + a\ell(u).$$

The Jacobian determinant is -1, the zero section is $\mathbb{P}^1$, and its normal bundle is $\mathcal{O}(1) \oplus \mathcal{O}(-3)$. The noncommutative deformation algebra of this neighborhood is $\operatorname{Jac}(G)$ by [2, Corollary 6.4]. Lemma 3.8 gives polynomial gluing data with the same completed Jacobi algebra. For the length-five stratum, we use the fifth family of Wemyss's construction theorem [6, Theorem 1.7]. The following proposition applies that theorem after normalizing the five-jet.

Proposition 3.9 (Length-five realization)**.** *Every potential in Theorem 3.3 has a geometric Jacobi algebra, realized by a smooth length-five* cE_8 *flop. Equivalently, the alternating-quartic stratum with nonzero pure fifth coefficient is realized, including all formal tails.*

Proof. The change $X = y + x^2$, $Y = x$ gives five-jet

$$\frac{X^3}{3} + \frac{(XY)^2}{2} - XY^4 - \frac{a_4}{5}Y^5.$$

At the level of five-jets, a shear $Y \mapsto Y + dX$, followed by the quadratic X correction that restores the quartic and by removal of cyclic XX terms, changes the coefficient B of XY^4 to $B + 5Cd$, where $C \neq 0$ is the pure Y^5 coefficient. Choose d to kill it. A quadratic Y correction then kills the residual $XYXY^2$ coefficient. Rescaling the variables and potential gives $X^3 + (XY)^2 + Y^5 + O_6$, the fifth row of the cited construction. These are finite coordinate changes; their unrecorded higher terms are retained in O_6.

Theorem 3.3 and Lemma 3.8 allow a polynomial representative with the same five-jet and identical Jacobi algebra. Wemyss's construction theorem produces the smooth length-five contraction. Complete its target at the contraction point and base change the source. The marked classification identifies its cDV type as cE_8. Transporting these Jacobi-algebra isomorphisms realizes the original algebra. □

4 Growth exclusions

Put $A_f = \mathrm{Jac}(f)$ and $J = \operatorname{rad} A_f$. A continuous invertible change of variables preserves the radical filtration and all dimensions $\dim_{\mathbb{C}} A_f/J^N$.

Theorem 4.1 (Zero-quartic exclusion)**.** *For every cyclic formal potential $f = x^3 + O_5$ and every $N \geq 1$,*

$$\dim_{\mathbb{C}} A_f/J^N \;\geq\; Z(N) := \sum_{k=0}^{\lfloor (N-1)/3 \rfloor} \binom{N-3k+1}{2} = \frac{N^3}{18} + O(N^2).$$

In particular $\operatorname{Jdim} A_f \geq 3$.

Theorem 4.2 (Both degenerate sextic axes)**.** *Let*

$$f = x^3 + y^4 + b(xy^2)^2 + c(xy)^3 + O_7, \qquad bc = 0,$$

with an arbitrary complete cyclic tail. Then $\operatorname{Jdim} A_f \geq 3$. *More precisely, after the weight-preserving normalization of Lemma 4.8, the following bounds hold for every positive integer N, and hence also hold for the original algebra:*

$$b = 0: \quad \dim_{\mathbb{C}} A_f/J^{3N} \geq 3\binom{N+2}{3},$$

$$c = 0: \quad \dim_{\mathbb{C}} A_f/J^{2N} \geq 2 \sum_{k=0}^{\lfloor (N-1)/2 \rfloor} \binom{N-2k+1}{2}.$$

The intersection $b = c = 0$ satisfies both bounds.

The next two subsections establish the quiver growth estimates and transfer those estimates to the sextic axes. The zero-quartic proof follows from the PBW and syzygy-lifting arguments in Subsections 4.4 and 4.5.

4.1 Two complete-quiver growth bounds

Let Q_q be the quiver on q vertices with exactly one arrow from each vertex to every vertex, including one loop at every vertex. Write $\mathfrak{a}$ for its arrow ideal. We need only $(q, m) = (3, 3)$ and $(2, 4)$.

The growth comparison starts from homogeneous seed potentials. Their finite quotients will be shown to be free over the formal ring of coefficient parameters at the seed. The following criterion reduces this freeness to vanishing of degree-minus-one cohomology. For a quiver potential, the Ginzburg differential graded algebra has the original arrows in cohomological degree zero, opposite arrows in degree minus one, and a vertex loop in degree minus two. Its degree-zero cohomology is the completed Jacobi algebra. The positive internal weights used below are separate from these cohomological degrees.

Lemma 4.3 (A finite flat-complex criterion)**.** *Let $(T, \mathfrak{n}, k)$ be a Noetherian local ring and let P be a nonpositive complex of flat T-modules. Suppose $H^0(P)$ is finite and $H^{-1}(P \otimes_T k) = 0$. Then $H^0(P)$ is finite free over T.*

Proof. Put $K = \operatorname{im}(P^{-1} \to P^0)$ and $B = H^0(P)$. If an element of $K \otimes k$ maps to zero in $P^0 \otimes k$, lift it to $P^{-1} \otimes k$. The lift is a cycle, so it is a boundary; its image in $K \otimes k$ is therefore zero. Thus $K \otimes k \to P^0 \otimes k$ is injective, so $\operatorname{Tor}_1^T(B, k) = 0$. A minimal surjection from a finite free module onto B has a finite kernel whose reduction modulo $\mathfrak{n}$ is zero. Nakayama's lemma then implies that this kernel is zero. □

Lemma 4.4 (Exact homogeneous seeds)**.** *There are homogeneous potentials W_3 of degree three on Q_3 and W_2 of degree four on Q_2 whose Ginzburg differential graded algebras have zero negative cohomology. Their Jacobi algebras have total Hilbert series*

$$H_3(t) = \frac{3}{(1-t)^3}, \qquad H_2(t) = \frac{2}{(1-t)^2(1-t^2)}.$$

For a seed of degree m, assign internal weight one to every arrow; opposite generators have weight $m-1$ and the degree-minus-two vertex generators have weight m.

Proof. For $q = 3$, let the cyclic group of order three act on $\mathbb{C}[X, Y, Z]$ with characters $0, 1, 2$. The McKay quiver is Q_3. The lift of the invariant cyclic potential $XYZ - XZY$ has Jacobi algebra $\mathbb{C}[X, Y, Z] \rtimes C_3$. Explicitly, with indices modulo three, take loops $a_i : i \to i$, arrows $b_i : i \to i+1$ and $c_i : i \to i-1$, and potential

$$W_3 = \sum_i (a_i b_i c_{i+1} - a_i c_i b_{i-1}).$$

Its derivatives are the three commutation relations, with vertex labels. The Ginzburg algebra is the cobar algebra of the exterior Koszul coalgebra on three generators, crossed with C_3. The Koszul resolution shows that the augmentation from this cobar algebra to the skew polynomial algebra is a quasi-isomorphism. The group has trivial character on the volume element, so the degree-minus-two generators are vertex loops, as required. PBW monomials give the first series.

For $q = 2$, take C_2 acting by $X \mapsto X$, $Y \mapsto -Y$. The lift of $\frac{1}{2}[X, Y]^2$ is a quartic potential on Q_2. The Jacobi algebra is

$$U(\mathfrak{h}_3) \rtimes C_2, \qquad [X, Y] = Z, \quad [X, Z] = [Y, Z] = 0,$$

where $\deg X = \deg Y = 1$, $\deg Z = 2$, and the involution sends (X, Y, Z) to $(X, -Y, -Z)$. PBW gives the second series. To prove negative acyclicity, stabilize the potential to

$$\widetilde{W} = Z[X, Y] - \tfrac{1}{2}Z^2.$$

Its Ginzburg algebra is the cobar construction of the Chevalley–Eilenberg coalgebra of $\mathfrak{h}_3$, crossed with C_2. The cobar augmentation to $U(\mathfrak{h}_3)$ is an equivariant quasi-isomorphism: the PBW filtration reduces this assertion to the exterior Koszul calculation for an abelian three-dimensional Lie algebra. The assigned internal weights are positive and the differential preserves them, so the assertion holds in each internal weight separately.

To remove the stabilizing pair, write the opposite generators as p, q, r, so that

$$dp = [Y, Z], \quad dq = [Z, X], \quad dr = [X, Y] - Z, \quad d\zeta = [X, p] + [Y, q] + [Z, r].$$

Set

$$Z' = Z - [X, Y], \quad p' = p + [Y, r], \quad q' = q + [r, X], \quad \zeta' = \zeta + r^2.$$

Then $dr = -Z'$, $dZ' = 0$, and

$$dp' = [Y, [X, Y]], \quad dq' = [[X, Y], X], \quad d\zeta' = [X, p'] + [Y, q'].$$

This is the quartic Ginzburg algebra together with a free contractible pair. The changes respect the involution and internal weights. This proves the asserted negative acyclicity. □

Proposition 4.5 (Uniform quiver bounds)**.** *Over any characteristic-zero extension of $\mathbb{C}$, the following hold. For every potential $W \in \mathfrak{a}^3$ on Q_3,*

$$\dim \mathrm{Jac}(W)/\mathfrak{a}^N \geq 3\binom{N+2}{3}.$$

For every potential $W \in \mathfrak{a}^4$ on Q_2,

$$\dim \mathrm{Jac}(W)/\mathfrak{a}^N \geq 2\sum_{k=0}^{\lfloor (N-1)/2 \rfloor} \binom{N-2k+1}{2}.$$

The potentials may be formal and need not be homogeneous.

Proof. Fix N and either pair (q, m). Only cyclic coefficients of degrees $m, \ldots, N$ affect the quotient. Vary all these coefficients independently around the seed W_q over a formal power-series ring T. In the relative Ginzburg complex assign weights $1, m-1, m$ to the arrows, opposites and vertex generators, respectively. The differential never lowers weight. Quotient the entire complex by weights at least N. Every resulting term is a finite free T-module, and the special complex has zero degree-minus-one cohomology by Lemma 4.4. The degree-zero cohomology of the relative complex after the weight cutoff is exactly the truncated relative Jacobi algebra. Lemma 4.3 therefore shows that the truncated relative Jacobi algebra is free over T with the seed's truncated Hilbert dimension.

The differential matrix into degree zero has polynomial entries in the finitely many potential coefficients. If its generic rank is r, every $(r+1)$-minor vanishes as a formal series at the seed, hence is identically the zero polynomial. At any coefficient point the rank is therefore at most r. Consequently every specialization has quotient dimension at least the seed dimension. Translation by the seed parametrizes the whole coefficient space. Summing the Hilbert coefficients in Lemma 4.4 gives the displayed formulas. Polynomial identities remain identities after extension of fields. Formal tails beyond degree N do not affect the quotient. □

4.2 Growth transfer by partial morsification

We compare finite quotients of the original Jacobi algebra with the split quiver algebra through modules over the deformation parameter. These modules retain every moving support point.

All roots of the split pure-power equation contribute to this comparison. Each root gives a vertex, and each component of the retained variable between two root spaces gives an arrow. The finite relative quotients below keep these vertices together when comparing the special and generic fibers.

Lemma 4.6 (Pure-power splitting and nilpotent quotients)**.** *Let*

$$f = P(u) + \frac{v^{q+1}}{q+1} + g(u, v),$$

where $P \in u^m\mathbb{C}[[u]]$ and every word of g contains at least m occurrences of u, with $m \geq 3$. Suppose that, for each fixed number of u's, only finitely many words of g occur; the same conclusion holds if the substitutions below converge in the corresponding adic topology. Put $A = \mathrm{Jac}(f)$, $I = (u)$ and $J = \operatorname{rad} A$. For each N there is a finite $\mathbb{C}[[s]]$-module M_N whose special fiber is A/I^N and whose generic fiber is

$$\mathrm{Jac}(W)/\mathfrak{a}^N$$

for a potential $W \in \mathfrak{a}^m$ on Q_q over $K = \mathbb{C}((s))$. In particular,

$$\dim_{\mathbb{C}} A/J^{qN} \;\geq\; \dim_{\mathbb{C}} A/I^N \;\geq\; \dim_K \mathrm{Jac}(W)/\mathfrak{a}^N.$$

Proof. Use the potential

$$F_s = f - s^q v$$

and let M_N be its relative Jacobi algebra, completed with respect to (s, u, v), modulo the Nth power of the ideal generated by u. Reduction modulo s gives A/I^N. Here $A/I \cong \mathbb{C}[[v]]/(v^q)$, so $J^q \subset I$ and $J^{qN} \subset I^N$. Each layer I^j/I^{j+1} is spanned by expressions $a_0 u a_1 \cdots u a_j$, where each a_i is taken modulo I. It is therefore finite-dimensional. Thus A/I^N is finite-dimensional. Topological Nakayama now makes M_N finite over $\mathbb{C}[[s]]$: a finite basis of the closed fiber lifts, and successive approximation gives coefficients in $\mathbb{C}[[s]]$; the total and s-adic topologies are cofinal because the closed radical is nilpotent. Reduction agrees with the stated closed-ideal quotient. Finite generation alone gives $\dim_{\mathbb{C}} M_N/sM_N \geq \dim_K M_N \otimes K$.

Over K the equation at $u = 0$ is $v^q - s^q = 0$, with distinct roots $\lambda_i = s\zeta_q^i$. Work around all of them simultaneously. Hensel idempotents for v give $e_1, \ldots, e_q$, and the completed free algebra at the semisimple quotient $K[v]/(v^q - s^q)$ has the presentation whose generators are

$$u_{ij} = e_i u e_j \quad (1 \leq i, j \leq q), \qquad t_i = e_i(v - \lambda_i)e_i.$$

It is the completed path algebra with the q^2 arrows u_{ij} and q additional loops t_i. This presentation also follows directly from the universal property: the inverse sends u to $\sum u_{ij}$, sends v to $\sum(\lambda_i e_i + t_i)$, and recovers the spectral idempotents by Hensel's construction.

In these coordinates the pure v-potential has a nondegenerate quadratic part in the variables t_i, since

$$(v^q - s^q)'|_{v=\lambda_i} = q\lambda_i^{q-1} \neq 0.$$

All other terms contain at least m u-arrows. The noncommutative splitting lemma, or successive completion of the squares in the t_i's, eliminates these loops while fixing the u_{ij}. The critical values satisfy $t_i \in (u_{ij})^m$. The reduced potential W therefore has no terms of arrow degree less than m.

The quiver derivatives are the blocks of $\partial_u F_s$ and the diagonal blocks of $\partial_v F_s$. Modulo them the cyclic identity gives $[v, \partial_v F_s] = 0$. On an off-diagonal block the operator

$$z \longmapsto (\lambda_i + t_i)z - z(\lambda_j + t_j)$$

is invertible for $i \neq j$: its constant part is multiplication by the nonzero scalar $\lambda_i - \lambda_j$, and the remaining operator is adic-nilpotent. Hence those off-diagonal derivatives vanish as well. The resulting Jacobi-algebra isomorphism preserves the ideal generated by the u_{ij}.

After imposing the Nth power of that ideal, the resulting quotient is finite-dimensional over K; completion at this semisimple quotient changes nothing. Thus the generic fiber of the original finite module M_N is exactly $\mathrm{Jac}(W)/\mathfrak{a}^N$. The comparison uses the completion at all roots λ_i, and therefore retains the full generic support. □

Remark 4.7. For the normalized potentials used below, the finite-word hypothesis is automatic. A cyclic word avoiding xx and yyy has at most twice as many y's as x's, and at most as many x's as y's. Bounding either letter count bounds its total length. Thus every finite nilpotent quotient uses only finitely many coefficients of the original formal potential.

4.3 Proof for the degenerate sextic axes

Lemma 4.8 (Weighted normalization)**.** *Give x, y weights $4, 3$. Every $x^3 + y^4 + O_5$ is cyclically right-equivalent to a potential whose higher necklaces avoid cyclic xx and cyclic yyy. Below weight 24 its only possible higher necklaces are*

$$a\,xyxy^2 \quad (\mathrm{wt}\,17), \qquad b(xy^2)^2 \quad (\mathrm{wt}\,20), \qquad c(xy)^3 \quad (\mathrm{wt}\,21).$$

For $x^3 + y^4 + b(xy^2)^2 + c(xy)^3 + O_7$, the normalization leaves b, c unchanged and creates no weight-17 term.

Proof. Lemma 2.15 gives the general normalization and convergence. In the special sextic-axis input there are no earlier bad terms capable of changing the displayed coefficients; the ordinary O_7 terms of weights $21, 22, 23$ are bad and their removal does not change any good necklace at those weights. The remaining good tail has ordinary order at least seven. □

Proof of Theorem 4.2. Apply Lemma 4.8. If $b = 0$, every mixed necklace contains at least three x's. Indeed, a good necklace with two x's has at most four y's, so has length at most six. The only such sextic necklace is $(xy^2)^2$, which is absent; the necklace of degree five and weight seventeen is also absent. Consequently, after rescaling,

$$f = \frac{x^3}{3} + \frac{y^4}{4} + g, \qquad \text{every word of } g \text{ contains at least three } x\text{'s.}$$

Use Lemma 4.6 with retained variable $u = x$, split variable $v = y$, and $(q, m) = (3, 3)$. Proposition 4.5 gives

$$\dim_{\mathbb{C}} A/J^{3N} \geq 3\binom{N+2}{3}.$$

If $c = 0$, every higher good necklace contains at least four y's: any good necklace with at most three y's has at most three x's, hence has length at most six; the only possible sextic necklace is $(xy)^3$, which is now absent. Thus retain $u = y$, split $v = x$, and use $(q, m) = (2, 4)$. This gives

$$\dim_{\mathbb{C}} A/J^{2N} \geq 2 \sum_{k=0}^{\lfloor (N-1)/2 \rfloor} \binom{N-2k+1}{2}.$$

Both right sides grow cubically. If both coefficients vanish, both arguments apply. □

4.4 PBW and syzygy lifting for the zero-quartic stratum

The zero-quartic bound comes from an associated-graded calculation on a dense open set of fifth jets. The finite certificate identifies the homogeneous relations and their syzygies. Exact lifts of those syzygies will preserve the associated graded algebra for every higher formal tail at those jets; semicontinuity will then give the bound at the remaining fifth jets.

Put $R = \mathbb{Q}\langle x, y\rangle$ with ordinary grading, and define

$$U = yxy^2 + y^2xy, \qquad R_0 = x^2, \qquad R_1 = y^4 + xy^2x + yxyx + xyxy.$$

Let $M = RpR \oplus RqR$, with generator weights $2, 4$, and let $d_0 : M \to R$ send p to R_0 and q to R_1.

Lemma 4.9 (Finite rational syzygy certificate)**.** *The kernel of d_0 is generated as an R-bimodule by*

$$\begin{aligned} s_1 &= xp - px, \\ \sigma &= x(yq - qy) + (yq - qy)x - Up + pU, \end{aligned}$$

and all Peiffer identities

$$R_i a \theta_j - \theta_i a R_j, \qquad i, j \in \{0, 1\}, \quad a \in R, \quad \theta_0 = p,\ \theta_1 = q.$$

Moreover

$$H_{R/(R_0,R_1)}(t) = \frac{1}{(1-t)^2(1-t^3)}.$$

Proof. Use degree-lexicographic order with $x < y$. A homogeneous Gröbner basis is

$$\begin{aligned} &x^2, \\ &y^4 + yxyx + xy^2x + xyxy, \\ &y^3xy^2x + y^3xyxy - yxyxy^3 - xy^2xy^3, \\ &y^3xyx + y^2xy^2x + y^2xyxy - yxyxy^2 - xy^2xy^2 - xyxy^3, \\ &y^2xyx + yxy^2x - xy^2xy - xyxy^2. \end{aligned}$$

The certificate retains the fourth element as a redundant generator. The accompanying standard-library Python verifier checks that all five elements are combinations of the two original relations and checks every overlap and inclusion; there are seventeen in total. After substitution back

into the original free bimodule, eleven critical syzygies are nonzero. Their weights and numbers are

weight	3	6	7	8	9	10
number	1	1	1	4	3	1.

The JSON certificate expresses these eleven syzygies using thirty-one explicit rational multiples of s_1, σ, and Peiffer identities. The verifier checks these identities in the free relation bimodule over the polynomial free algebra, using exact rational coefficients.

Critical syzygies generate the full kernel as follows. In an identity among relations, compare reductions at the largest leading word. Intersecting occurrences give an overlap or an inclusion identity; disjoint occurrences give a Peiffer identity. Subtracting these comparisons lowers the largest word. Termination gives the assertion for every polynomial identity. The new Gröbner generators are replaced by their recorded expressions in terms of the original relations; their defining identities then vanish. Expanding a Peiffer identity for these expressions bilinearly gives a sum of Peiffer identities for the original generators. This is the usual confluence proof of critical syzygy generation and explains why the finite check controls every weight.

The minimal forbidden words are

$$xx, \qquad yyyy, \qquad yyxyx, \qquad yyyxyyx.$$

Their prefix automaton has ten states. The certificate supplies exact polynomial numerators $n_i(t)$ with common denominator $D(t) = (1+t)(1-t)^2(1-t^3)$ satisfying

$$n_i(t) = D(t) + t \sum_{i \to j} n_j(t).$$

The numerator at the empty state is $1+t$. Since the automaton equations have a unique formal solution, these verified polynomial identities prove the asserted rational Hilbert series in every degree. □

Lemma 4.10 (Liftable syzygies and a closed initial ideal)**.** *Let r_i be filtered formal relations with homogeneous initial relations R_i. If every homogeneous syzygy of the R_i is a sum of homogeneous syzygies admitting exact syzygy lifts for the r_i with the same initial terms, then*

$$\operatorname{gr}(\widehat{R}/\overline{(r_i)}) \cong R/(R_i).$$

Proof. Take any nonzero element of the closed relation ideal, of order n, and represent it by relation multiples modulo order $n+1$. If the leading part of this representation cancels, it is a homogeneous syzygy of the R_i. Subtract an exact lifted syzygy, without changing the represented element. The least weight of the representation increases. Repeating this process finitely many times, up to weight n, gives a representation whose first nonzero image is a combination of the R_i in weight n. Thus every initial relation belongs to (R_i). The reverse inclusion is immediate. The compatible finite-cutoff arguments give the conclusion for the closed ideal. □

Proposition 4.11 (Generic fifth-jet normal form)**.** *For every cyclic $h \in O_6$, let*

$$f = \frac{x^3}{3} + xyxy^2 + \frac{y^5}{5} + h.$$

Then

$$\operatorname{gr}_J \operatorname{Jac}(f) \cong \frac{\mathbb{C}\langle x, y\rangle}{(x^2,\ y^4 + xy^2x + yxyx + xyxy)}.$$

In particular, $\operatorname{gr}_J \operatorname{Jac}(f)$ *has Hilbert series* $1/((1-t)^2(1-t^3))$.

Proof. The two relations are

$$r_x = x^2 + U + \partial_x h, \qquad r_y = R_1 + \partial_y h.$$

Assign weights $2, 4$ to the formal relation symbols p, q. The universal cyclic identity gives the exact syzygy

$$S_1 = [x, p] + [y, q], \qquad dS_1 = [x, r_x] + [y, r_y] = 0.$$

Its initial term is s_1 of weight three. There is a second exact syzygy

$$S_2 = xS_1 + S_1x - r_xp + pr_x.$$

Indeed $dS_2 = 0$. Cancelling its weight-four terms gives

$$S_2 = \sigma - (\partial_x h)p + p(\partial_x h).$$

Since $\partial_x h$ has order at least five, the last two terms have weight at least seven. Thus S_2 exactly lifts the weight-six syzygy σ. Every Peiffer identity lifts tautologically by replacing R_i with r_i. Lemma 4.9 therefore supplies every required syzygy lift, and Lemma 4.10 proves the associated-graded identification.

□

4.5 Semicontinuity for the zero-quartic stratum

Proof of Theorem 4.1. First remove the degree-five cyclic necklaces containing xx by an order-three correction of x. The remaining fifth jet is

$$\frac{x^3}{3} + Axyxy^2 + Bxy^4 + Cy^5.$$

For $C \neq 0$, applying the shear $y \mapsto y + tx$ and then removing the bad xx necklaces it creates changes the good coefficients to

$$A' = A + 2Bt + 5Ct^2, \qquad B' = B + 5Ct, \qquad C' = C.$$

Consequently $B^2 - 5AC$ is unchanged. On the dense open set

$$C \neq 0, \qquad B^2 - 5AC \neq 0,$$

choose $t = -B/(5C)$, and then rescale the two variables and the potential. The fifth jet becomes the one in Proposition 4.11. No quartic term is created. Every subsequent term remains an

unrestricted ordinary O_6 tail. These changes preserve the radical filtration, so the proposition computes the truncated dimensions on this whole dense open set, for every higher coefficient.

Fix N. The image of the closed derivative ideal in the finite word quotient modulo words of length N is the span of finitely many two-sided relation multiples. Its relation matrix is polynomial in finitely many potential coefficients. On the dense open set just described, its cokernel dimension equals the cumulative Hilbert function of the algebra in Proposition 4.11. Every larger rank minor vanishes there and hence identically. At an arbitrary fifth jet the cokernel dimension can only increase. Higher formal terms invisible at this cutoff do not change it. This proves the bound for every N and every zero-quartic formal potential, including degenerate fifth jets.

Finally

$$[t^n]\frac{1}{(1-t)^2(1-t^3)} = \#\{(a,b,c) \in \mathbb{Z}^3_{\geq 0} : a+b+3c = n\}.$$

Summing over $n < N$ gives $Z(N)$ and its cubic asymptotic. □

5 Proof of the realization theorem

5.1 The growth-one model

The positive Type E constructions leave only

$$\mathrm{Jac}(x^3 + xy^3)$$

as an infinite-dimensional Type E algebra, up to complete-algebra isomorphism. The two growth-exclusion theorems rule out both finite-dimensionality and growth one on the residual branches.

For the surviving model one can use the marked E_6 triangle with common central value

$$\beta^3 = w, \qquad \gamma^3 = -w, \qquad \delta^2 = -w/3, \qquad \beta+\gamma+\delta = 0.$$

The constant coefficients of the two cubic polynomials and the quadratic polynomial have nonzero first derivatives, so the source is regular. The opposite cubics give the persistent coefficient-two root locus, hence a divisor-to-curve crepant contraction. The three-direction potential for this marked E_6 triangle is $-\delta^3 + \beta^4/4 - \gamma^4/4$. Putting $x = \beta+\gamma$, $y = (\beta-\gamma)/2$ gives the cyclic expression

$$x^3 + xy^3 + \tfrac{1}{4}x^3y.$$

The weighted E_6 elimination in Theorem 2.13 may be performed compatibly with $x \mapsto -x$ and multiplication of the potential by -1. The potential above is odd in x, whereas the only possible pure-y residuals are even. They therefore do not occur, and the resulting normal form is $x^3 + xy^3$. This identifies the completed Jacobi algebra of the explicit divisorial construction.

Theorem 2.13 proves that every nonzero pure-series perturbation in this stratum is finite-dimensional. This leaves the displayed algebra as the unique growth-one Type E case.

5.2 Completing the quartic division

Proof of Theorem 1.3. Proposition 2.1 reduces the conjecture to $x^3 + O_4$. Proposition 2.3 gives the five quartic orbits. We apply the preceding constructions and exclusions to each orbit.

If $\Delta \neq 0$, Theorem 2.13 gives a marked length-three realization, including its classification into finite dimension and growth one.

In the alternating-quartic row, Theorem 3.1 forces a nonzero pure fifth coefficient. Corollary 3.2 normalizes the five-jet. The inverse of the change $X = y + x^2$, $Y = x$ preserves ordinary order, so the normalized potential becomes $F_p + h$, with $p(u) = u^4$ and $h \in O_6$. Theorem 3.3 proves finiteness for the entire tail, and Proposition 3.9 gives the length-five realization.

For the pure y^4 row, Lemma 4.8 gives the three coefficients of weights $17, 20, 21$. A nonzero weight-17 coefficient is covered by Theorem 2.16. If that coefficient is zero, Theorem 4.2 forces the product of the sextic coefficients to be nonzero; then Theorem 2.17 applies. Finally, Theorem 4.1 excludes the zero-quartic row.

Generic surface types. For the explicit positive marked constructions above, varying a hyperplane inside their fixed threefold deforms the distinguished partial-resolution surface while preserving its scheme fiber. Its generic Du Val graph is a subgraph of the distinguished graph, with the same retained-cycle multiplicity. For the length-three E_6 family no proper subgraph has multiplicity three, so its generic type is cE_6, including the explicit divisorial member. For the small primitive constructions of lengths four, five, and six, the invariant-theory tests of [4, Section 8] give generic types cE_7, cE_8, cE_8, respectively. These are the surface types recorded in the table below.

The resulting exhaustive table under Jdim ≤ 1 is

normalized branch	construction	generic surface type
$\Delta \neq 0$	marked length 3	cE_6
$(xy)^2$ quartic, nonzero pure fifth	marked length 5	cE_8
y^4 quartic, $a_{17} \neq 0$	marked length 4	cE_7
y^4 quartic, $a_{17} = 0,\ b_{20}c_{21} \neq 0$	marked length 6	cE_8

The growth bounds exclude every other branch.

Lemma 2.5 identifies the normalized basic contraction algebras of the marked-arc constructions. In the length-five construction, Lemma 3.8 supplies the polynomial representative used in Proposition 3.9. The coordinate changes and scalar multiples preserve the isomorphism class of the completed Jacobi algebra, so the constructions realize the original algebras.

Proposition 2.1 includes the finite and growth-one A/D cases. Each surviving branch therefore satisfies Definition 1.1, with the finite-dimensional case flopping and the growth-one case divisorial. □

6 Computational certificates

From the paper directory, run

```
cd tools/numerical_realisation_certificates
python3 verify_zero_quartic.py
```

with the adjacent `zero_quartic_certificate.json`. Only the Python standard library is needed. The verifier checks the five original-ideal identities, all seventeen ambiguities, all eleven free-bimodule syzygy decompositions, and the exact rational Hilbert series. The syzygy decompositions have thirty-one summands in total. The automaton certificate verifies polynomial identities for the rational Hilbert series.

The certificate establishes Lemma 4.9. Lemma 4.10 extends the syzygies to formal relations. Subsections 4.1 and 4.2 give the quiver estimates and their transfer, and Section 5 combines them with the geometric constructions.

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For this reason, I do not wish the existence or priority date of this manuscript to discourage a genuinely independent human derivation of these results. If a subsequent work obtains the same results by an independently developed, entirely human mathematical argument, I make no claim to special priority or intellectual credit for the corresponding proof ideas beyond the ordinary bibliographic acknowledgement of prior or concurrent results.

If, however, a subsequent argument makes substantive use of the results, constructions, or proof strategies contained in this manuscript, ordinary scholarly citation remains appropriate regardless of the tools used. In particular, when closely related arguments arise in AI-assisted work, I would appreciate explicit attribution, since the provenance of ideas generated through overlapping language-model systems can otherwise be difficult to determine.

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